

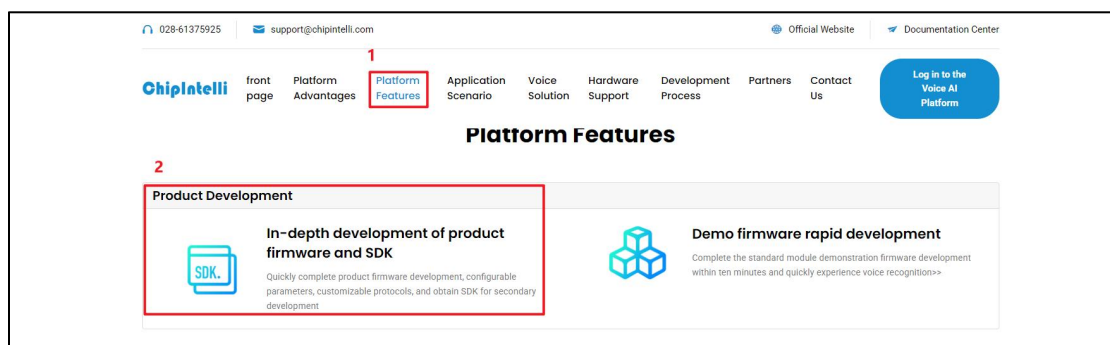
WonderEcho Pro Firmware Development

In this lesson, you'll learn how to create and flash firmware for the WonderEcho Pro module.

1. Firmware Creation and Flashing

We'll use “HIWONDER” as an example to demonstrate the firmware creation process. The same procedure also applies to creating firmware with Chinese wake words. Note: English wake words must be entered in all uppercase letters to function properly.

- 1) Open the official firmware creation website:
<https://aiplatform.chipintelli.com/home/index.html>; The platform interface is in Chinese. If needed, use your browser's translation feature to switch to English for easier navigation.
- 2) First, visit the ChipIntelli Voice AI Platform by clicking on “Platform Features” in the top menu, and then select “In-depth development of Product Firmware & SDK” under the Product Development section.



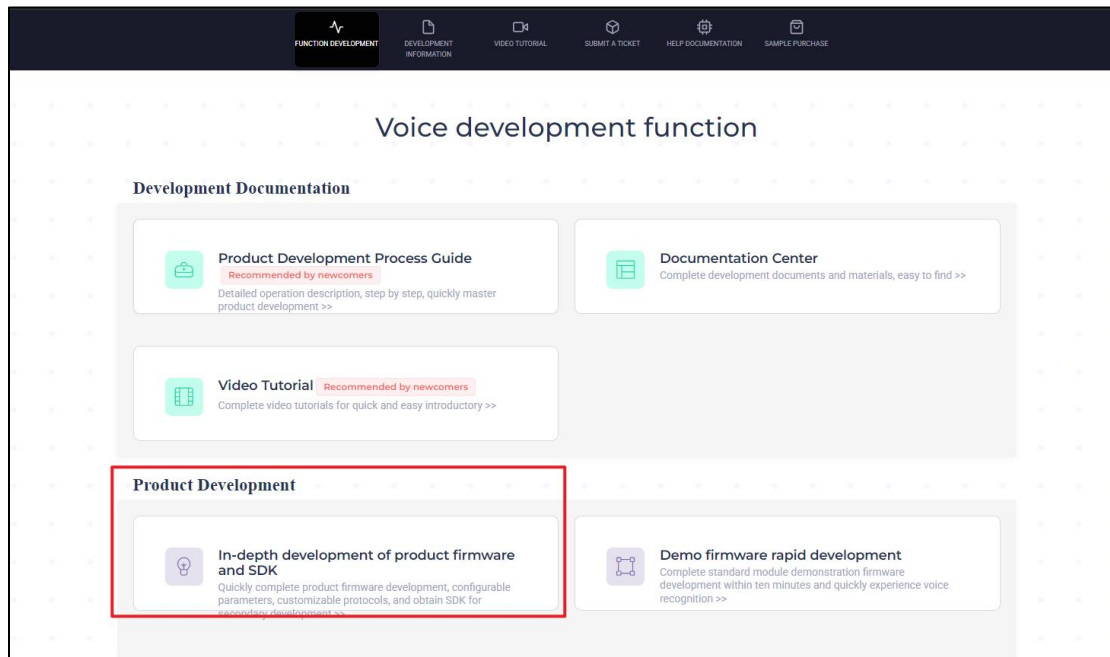
- 3) You will be prompted to log in. (Note: Currently, registration is only available for mainland China mobile numbers. For international users,

we provide a shared account:

Username: 18033085495

Password: hiwonder)

After successfully logging in, click “In-depth development of Product Firmware & SDK” again to access the development section.



- 4) To create a new product, click the option in the left sidebar. You will be asked to provide a Product Name and a Description, both of which can be customized. For the other fields, follow the options shown in the red box. Specifically, choose Product Type: 通用 (General) → 智能中控 (Smart Central Control). After completing these steps, click Create.

Product Information

* Product Name

hiwonder

* Application

单麦离线串口透传方案

Solution

* Product Type

CI13LC scheme

Single microphone offline serial port transparent transmission solution

Offline voice broadcast solution

Heater_Single Microphone Offline Slave Solution

Lighting_Two-way ceiling lamp offline master control solution

Range hood_single microphone offline slave solution

Yuba_single microphone offline slave solution

Electric curtain_single microphone offline slave solution

* SDK version

* Chip Model

* describe

create

Cancel

Product Information

* Product Name

hiwonder

* Application

单麦离线串口透传方案

Solution

* Product Type

智能中控

New categories

General

Voice Air Conditioner Companion

Intelligent central control

Kitchen and Bathroom

living room

bedroom

balcony

* SDK version

* Chip Model

* describe

create

Cancel

Product Information

* Product Name

hiwonder

* Application

单麦离线串口透传方案

Solution

* Product Type

智能中控

New categories

* SDK version

1.12.16

* Chip Model

CI1302

* describe

hello world

11 / 200

create

Cancel

- 5) Next, you will need to set the project language. Since we are configuring for English voice recognition, select English under Language Type. If you were configuring for Chinese recognition, you would select Chinese for both Voice Type and Acoustic Model. Once you've selected the appropriate options, click Continue to proceed.

Version Name

V1.0.0

Language

英文

Type

Chinese

Acoustic

English

Module board

Model

selection

continue

Version Name: V1.0.0

Language: 英文

Type: Selecting an

Acoustic Module board Model selection

English Standard General V2_1.3M_V00570

English Standard General V3_1.6M_V00614

English Standard General V3_1.6M_V00618_Applicable to sdk1.6.2 and above

English Standard General V4_1.1M_V00627_Applicable to sdk1.6.2 and above

English lamp universal V1_0.6M_V00686_Applicable to sdk1.6.2 and above

English general model V1_1.1M_V00916_Applicable to sdk1.6.2 and above

continue

Version Name: V1.0.0

Language: 英文

Type: Selecting an 英文通用模型V1_1.1M_V00916_适用sdk1.6.2及以上版本

Acoustic Module board Model selection CI-D02GS02S

continue

- 6) In the firmware configuration interface, you will make some important changes to the settings. Start by enabling the Echo Cancellation feature under Algorithm Settings.

Algorithm parameters	parameter	illustrate	Options
	Echo Cancellation	Applicable to scenarios that require voice interruption (such as music broadcast). You need to confirm that the module hardware supports this function before you can turn it on.	close ☒ open
Hardware Parameters	parameter	illustrate	Options
	Core 1.1V power supply	External DCDC or internal LDO power supply	内部
	Crystal Oscillator Source	Internal RC or external crystal source optional	内部RC

- 7) Under Hardware Settings, set the oscillator source to Internal RC and disable the Baud Rate Calibration function.

Hardware Parameters	parameter	illustrate	Options
	Core 1.1V power supply	External DCDC or internal LDO power supply	内部
	Crystal Oscillator Source	Internal RC or external crystal source optional	external internal
	Baud rate calibration function	Using the baud rate calibration function when using the internal RC	close ☐ open

Hardware Parameters	parameter	illustrate	Options
	Core 1.1V power supply	External DCDC or internal LDO power supply	内部
	Crystal Oscillator Source	Internal RC or external crystal source optional	内部RC
	Baud rate calibration function	Using the baud rate calibration function when using the internal RC	Internal RC External crystal

Hardware Parameters	parameter	illustrate	Options
	Core 1.1V power supply	External DCDC or internal LDO power supply	内部
	Crystal Oscillator Source	Internal RC or external crystal source optional	内部RC
	Baud rate calibration function	Using the baud rate calibration function when using the internal RC	close ☐ open

8) In the UART Print Settings, configure UART1 to Open-Drain Mode, supporting external 5V pull-up.

Print serial port configuration	parameter	illustrate	Options
	Print Serial Port	Print debug information serial port (baud rate 921600), cannot be the same as the communication serial port	UART1
	UART1 level	Configuration pin open-drain function, support external pull-up 5V	close ☒ open

9) For Communication UART Settings, set the Baud Rate to 115200 and configure UART0 to Open-Drain Mode, also supporting external 5V pull-up. After adjusting these settings, click Continue to move forward.

Communication serial port configuration	parameter	illustrate	Options
	Serial communication	When enabled, the voice module can communicate with the main control CPU through the serial port. When disabled, it cannot communicate with the main control CPU.	close ☒ open
	Communication serial port	Communication serial port name	UART0
	Communication serial port baud rate	The baud rate of the UART used for communication	115200
	Serial port protocol version	Customized serial port protocol, you can customize any protocol, the length of each protocol is less than 20 bytes. When the voice chip receives the protocol packet, it determines that the packet has been received by receiving the timeout (0.34ms). Therefore, when the main control sends the protocol packet, it needs to ensure that the time interval between packets is more than 0.34ms to ensure normal communication.	自定义协议
	UART0 level	Configuration pin open-drain function, support external pull-up 5V	close ☒ open

10) The next step involves configuring the command words. You will need to choose a voice output for the wake-up command. For this example, select “Dane – English Male Voice” as the output voice. (If you're using Chinese as the language, you may choose the voice option “Chengcheng – Standard Boy Ver.1”. The selection between Chinese and English voices is determined by the language settings configured in Step 5.)

Tone selection

English-Adult Female Voice

English-Adult Male Voice

English-Girl Voice

David-English Male Voice Ver.3

Daniel-English Male Voice Ver.3

James-English Male Voice Ver.3

John-English Male Voice Ver.3

Dane-English Male Voice Ver.1

Allen-English Male Voice Ver.1

音色选择

中文-成年女声

中文-成年男声

中文-女童声

中文-男声

中文-情感女声

中文-情感男声

程程-标准男声 Ver.1

11) Next, upload the command word attachment by locating the file named '(hello hiwonder) Command Word Broadcast Protocol List V3_English Template' and dragging it directly onto the webpage.

Note: If you wish to change the wake word, open the spreadsheet and replace the default wake word HIWONDER with your preferred word.

Keep in mind that the English version of the firmware only supports English wake words, and mixed-language wake words are not allowed. Additionally, English wake words must be written in uppercase letters to function properly.

语义标签	命令词	功能类型	播报语句	播报模式	发送协议	接收协议
1	GREETING	欢迎语	Welcome	被	AA 55 01 00 FB	AA 55 01 00 FB
2	SLEEP	休息语	I'm going to take a break	被	AA 55 02 00 FB	AA 55 02 00 FB
3	HELLO-HIWONDER	唤醒词	I'm here	被	AA 55 03 00 FB	AA 55 03 00 FB

12) After uploading the file, you will see the command data populated in the table below.

新增行	删除选中	格式检查: ☒ 开	*发送协议、接收协议、置信度可根据实际应用进行修改					
☐	* ☒ 语义标签	* ☒ 命令词	* ☒ 功能类型	* ☒ 播报语句	* ☒ 播报模式	* ☒ 发送协议	* ☒ 接收协议	☒ 置信度阈值
☐	1	GREETING	欢迎语	Welcome	被	AA 55 01 0 0 FB	AA 55 01 0 0 FB	35
☐	2	HAVE-A-REST	休息语	I'm going to take a break	被	AA 55 02 0 0 FB	AA 55 02 0 0 FB	35
☐	3	HIWONDER	唤醒词	I'm here	主	AA 55 03 0 0 FB	AA 55 03 0 0 FB	35
☐	4	TURN-UP-VOLUME	增大音量	Volume increase	主	AA 55 04 0 0 FB	AA 55 04 0 0 FB	35
☐	5	TURN-DOWN-VOLUME	减小音量	Volume decrease	主	AA 55 05 0 0 FB	AA 55 05 0 0 FB	35
☐	6	MAXIMUM-VOLUME	最大音量	Maximum volume	主	AA 55 06 0 0 FB	AA 55 06 0 0 FB	35

- 13) Click “Submit” and wait a few minutes for the firmware generation process to complete. Once finished, click “Download” to obtain your customized firmware.

SDK

☐ SDK Options

Download

Wake-up word

close

open

switch

Self-learning

close

open

function

Previous

save

Submit Now

Version Name	Version Numbering	Chip Model	Language Model	Creation time	Current Process	Feedback Description	operate
V1.0.0	sfw2025042515403971258201	C1302	V00916	April 25, 2025 00:40:39	Submit	in progress	delete inherit Check No downloads yet

Version Name	Version Numbering	Chip Model	Language Model	Creation time	Current Process	Feedback Description	operate
V1.0.0	sfw2025042515403971258201	C1302	V00916	April 25, 2025 00:40:39	Complete	OK	delete inherit Check Download File